VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

AGILE DEVELOPMENT PROCESS AND DEVOPS LAB

(LAB ASSIGNMENT : 1)

NAME : SIVASHREE.PL

REG NO : 23MIS0117

SLOT : C1+TC1(L49+L50)

CODE : ISWE406P

Use appropriate Git Commands to Execute the following case studies in VS Code

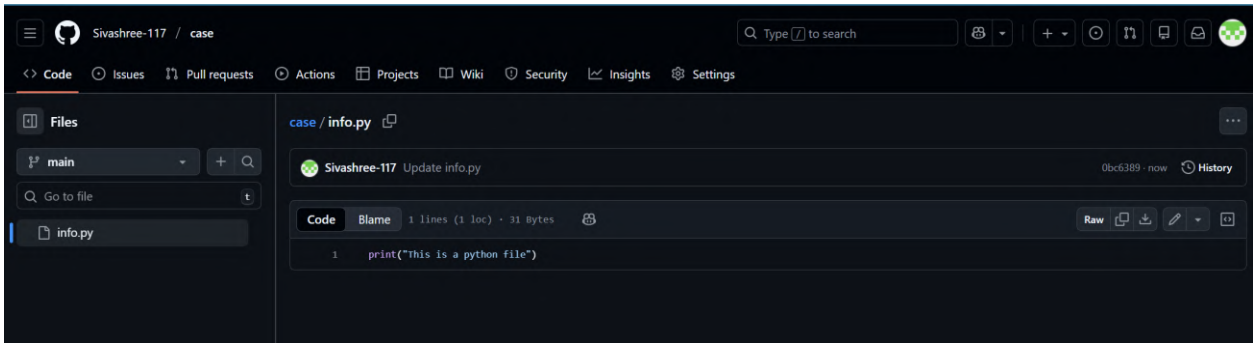
Case Study 1: New Project Version Control Setup

Scenario: You are assigned to develop a Library Management System. Before coding, your team decides to enable version control.

Tasks:

- Create a project folder and open it in VS Code

(New repo folder in git)

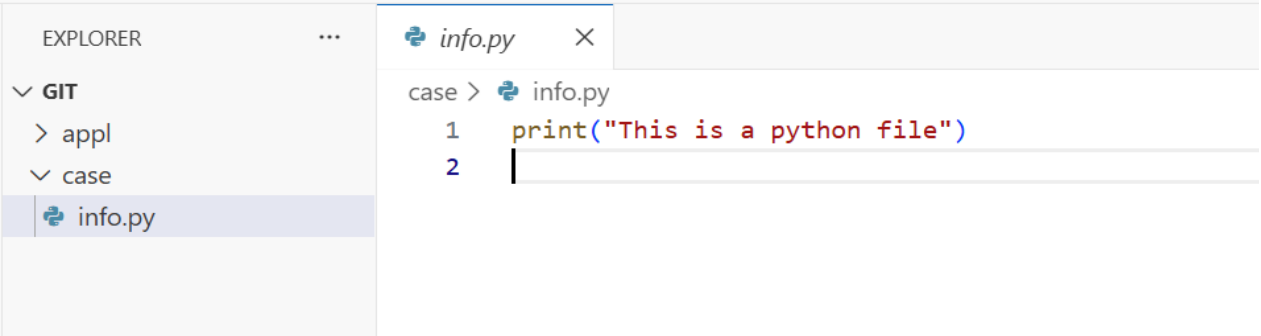


(Clone repo in vs code)

➔ **git clone url**

```
OUTPUT    DEBUG CONSOLE    PROBLEMS    TERMINAL    PORTS
PS C:\Users\sivas\OneDrive\Desktop\Git> git clone https://github.com/Sivashree-117/case.git
Cloning into 'case'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
PS C:\Users\sivas\OneDrive\Desktop\Git> cd case
```

(Git files copied to local PC)



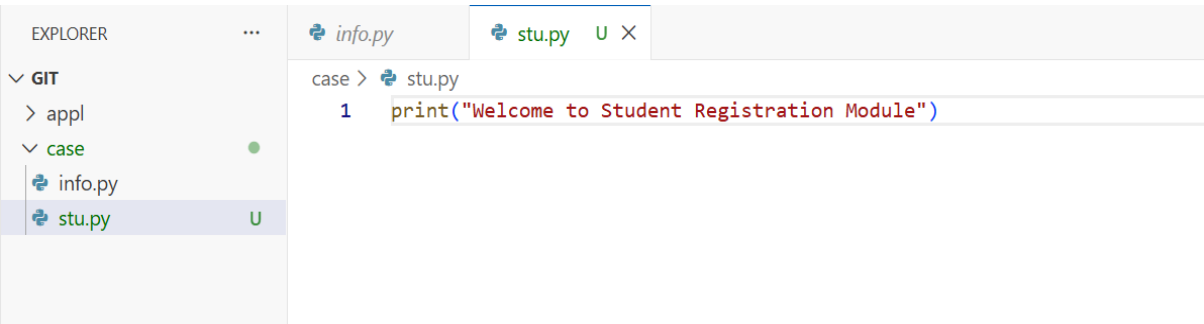
- Initialize a Git repository

→ **git init**

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git init
Reinitialized existing Git repository in C:/Users/sivas/OneDrive/Desktop/Git/case/.git/
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
```

- Create a Python file and track it using Git

(A new python file in created in vs code)



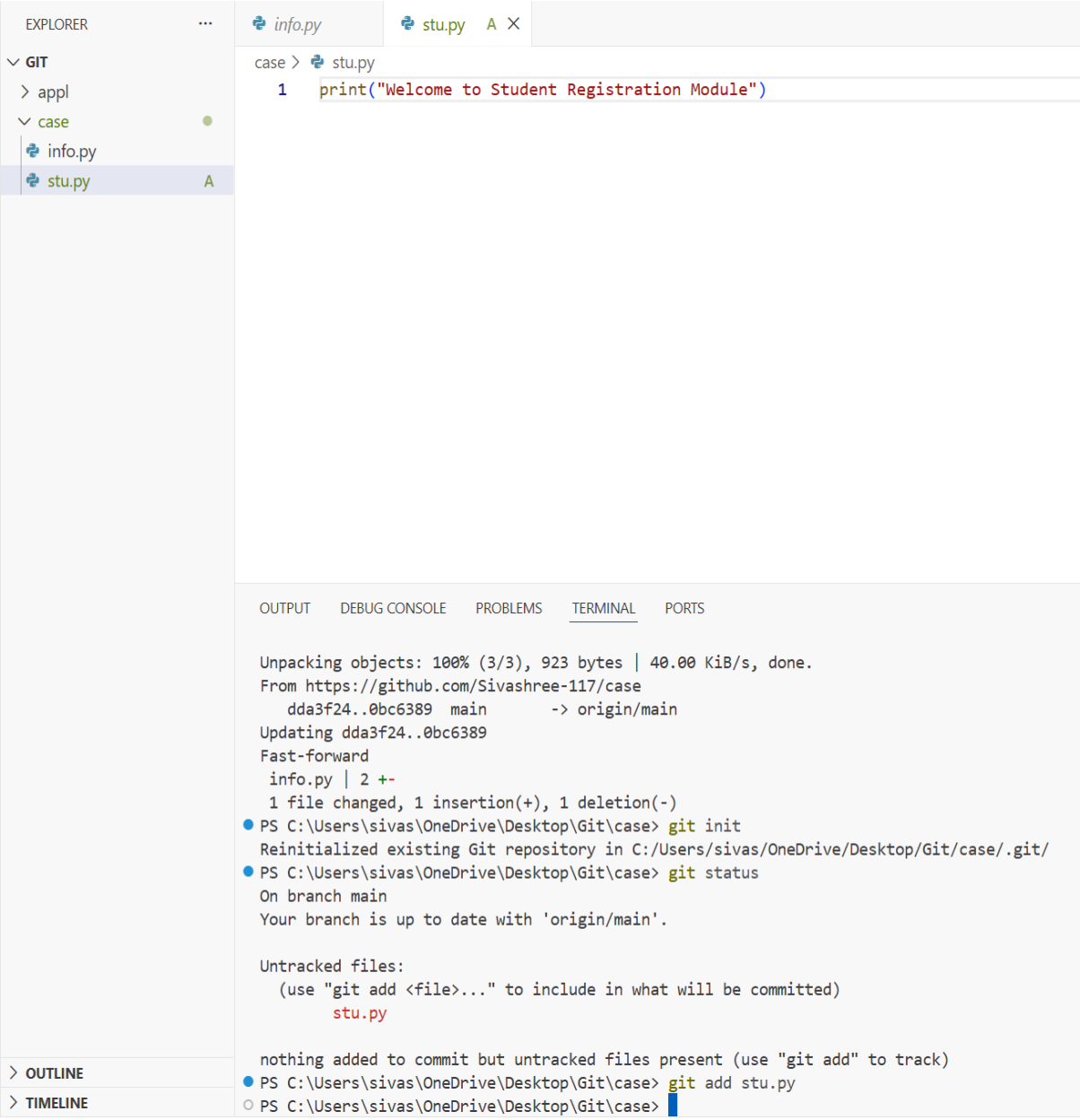
(New file is shown in git status because it is not yet added to main)

➔ **git status**

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        stu.py

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
```

(New Git file is added to main)

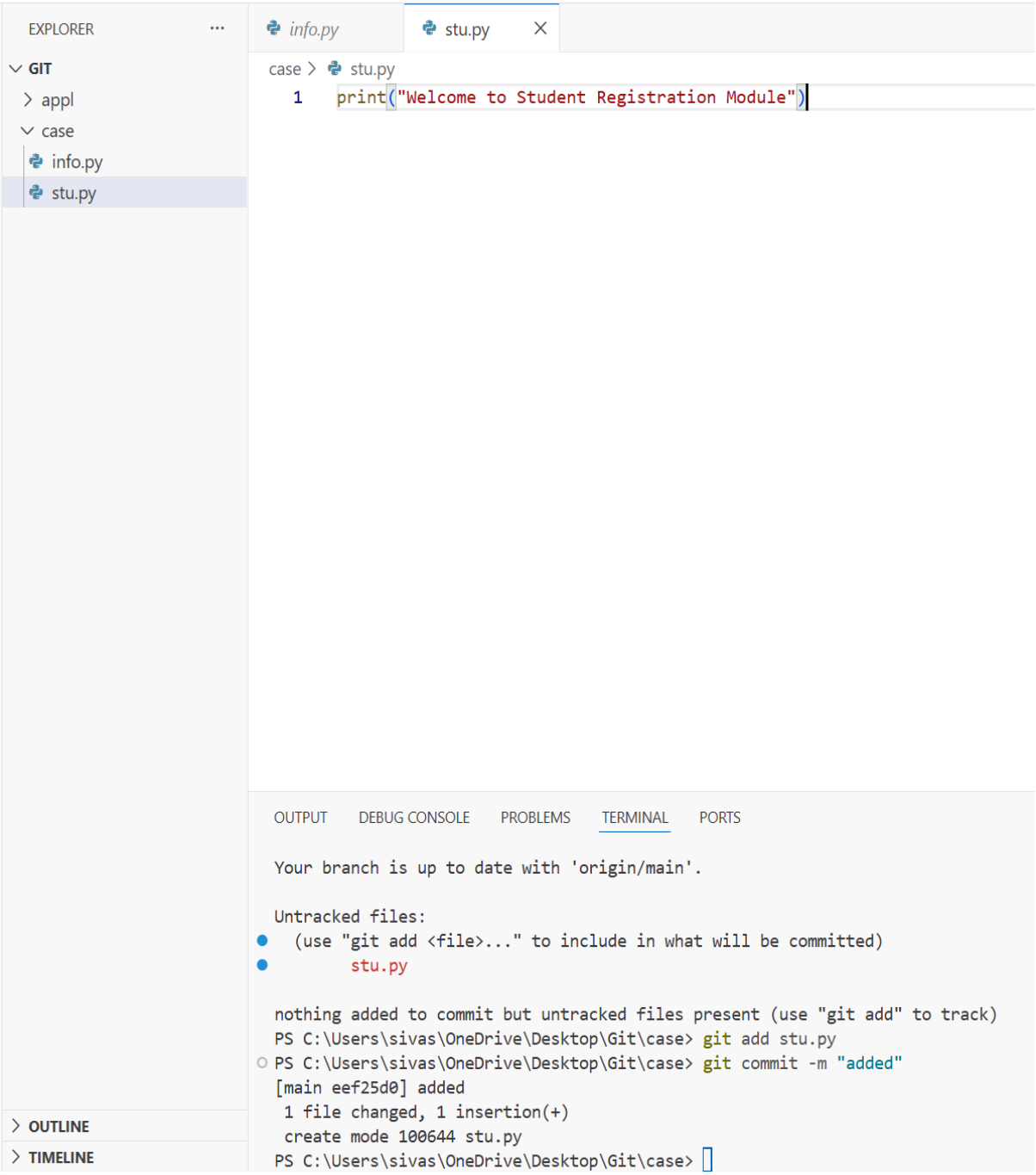
➔ **git add filename**



- Commit the initial version

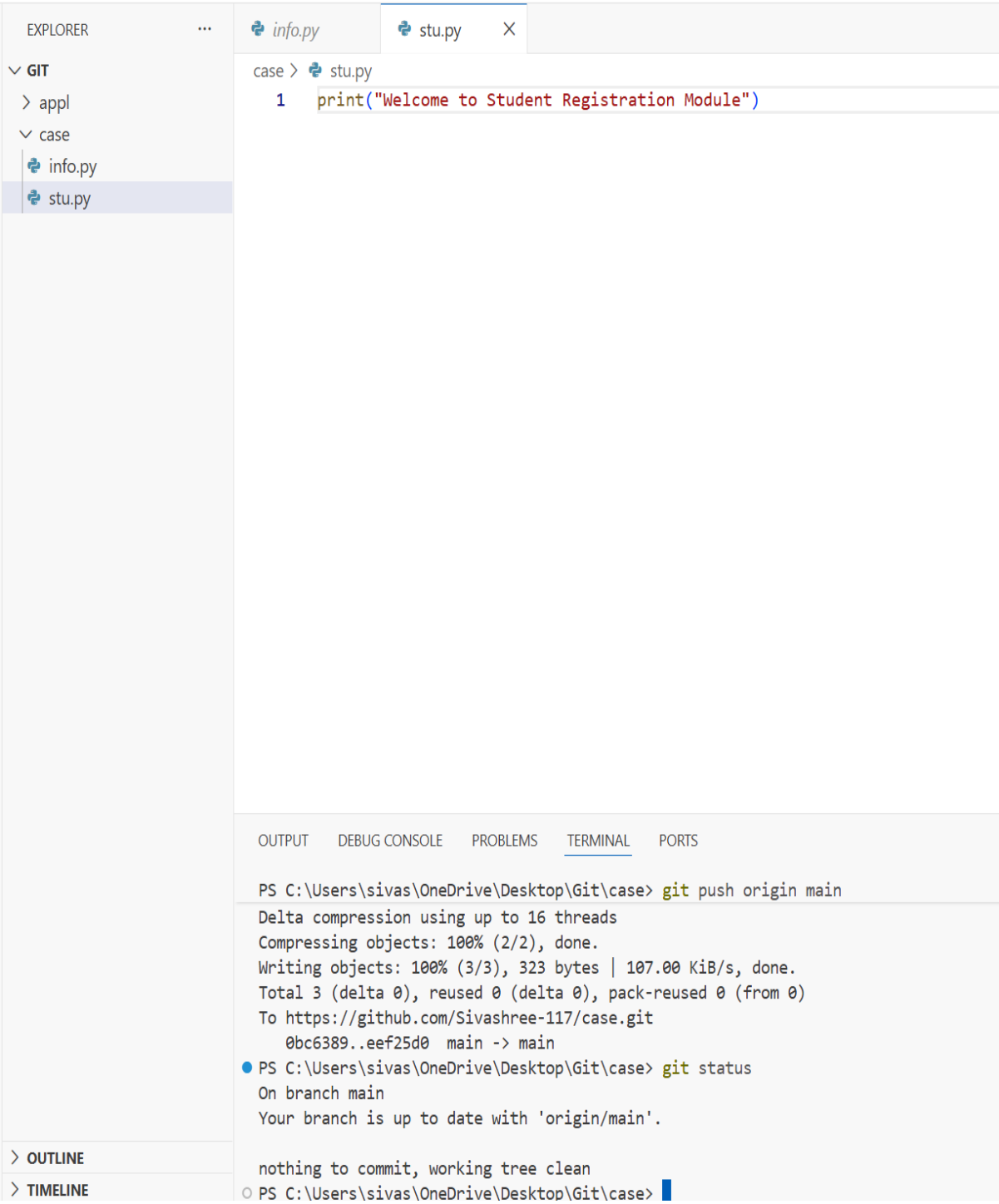
(After commiting file will be added to github)

➔ **git commit -m “comment”**

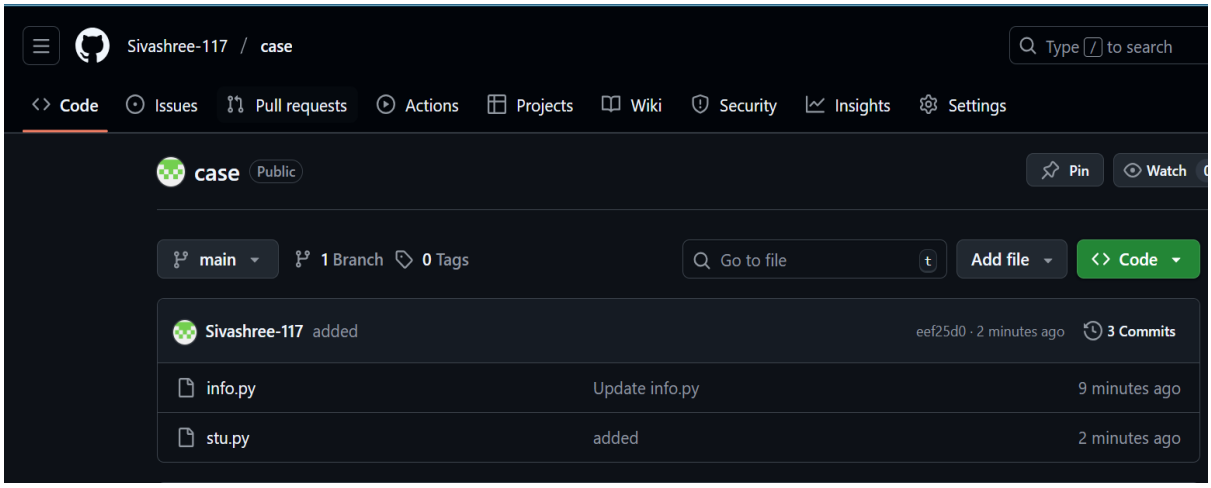


(This file is now pushed to hub)

➔ **git push origin main**



(New file is visible in github repository)



Case Study 2: Tracking Daily Code Changes

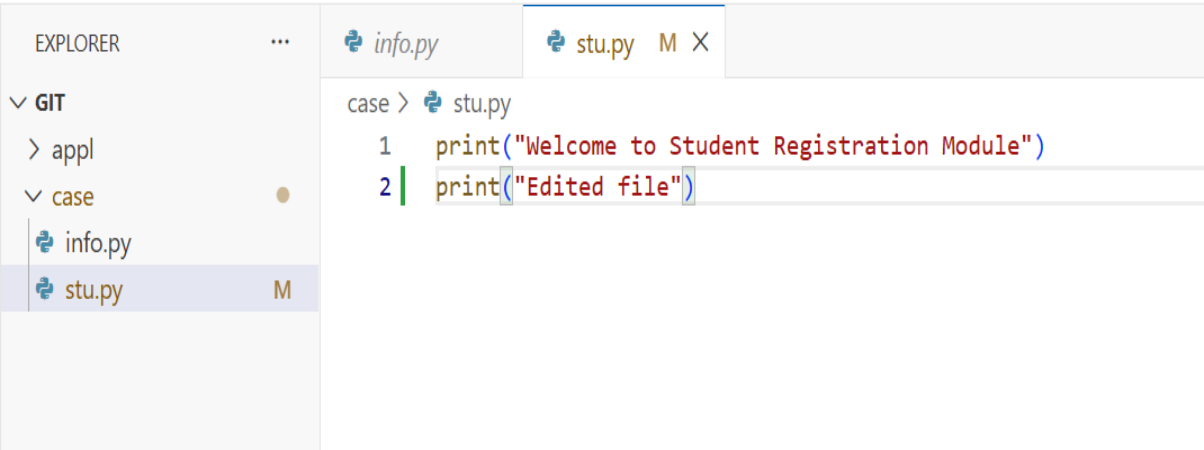
Scenario: You are working on a Student Registration Module. You make small updates daily and

need to record them.

Tasks:

- Modify an existing file

(Edit this file in vs code)



- Check file differences

(See diference from git and vs code)

➔ **git diff**

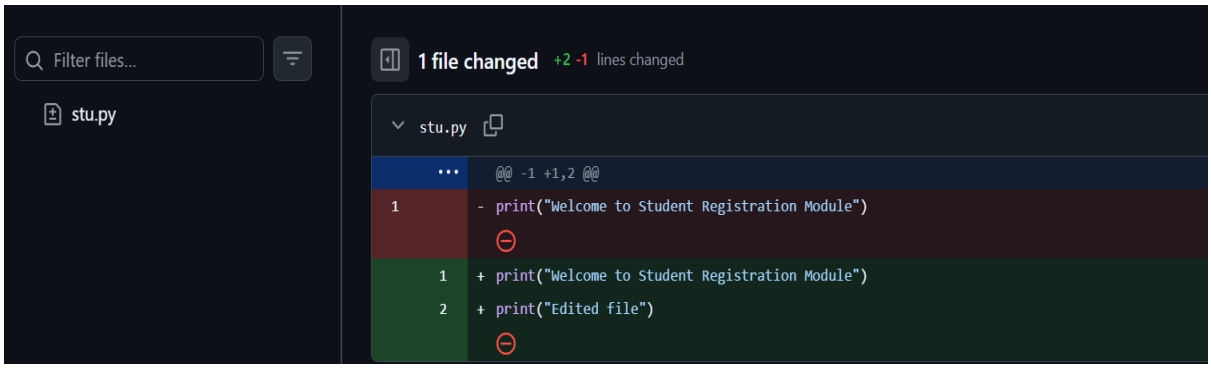
```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git diff
diff --git a/stu.py b/stu.py
index 3005012..e929a12 100644
--- a/stu.py
+++ b/stu.py
@@ -1 +1,2 @@
-print("Welcome to Student Registration Module")
\ No newline at end of file
+print("Welcome to Student Registration Module")
+print("Edited file")
\ No newline at end of file
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
```

- Stage and commit changes

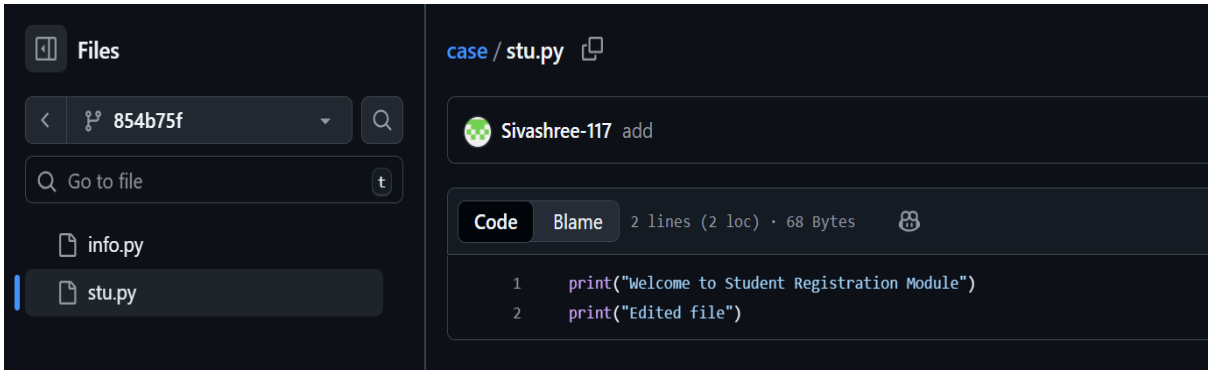
(Now add , commit and push it to main)

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git add stu.py
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git commit -m "add"
[main 854b75f] add
 1 file changed, 2 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 338 bytes | 112.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/case.git
   eef25d0..854b75f  main -> main
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
```

(Difference in old file and edited file)



(This is the edited file)



- View commit history

→ **git log**

```
EXPLORER ...
GIT
  appl
  case
    info.py
    stu.py
OUTLINE
TIMELINE

OUTPUT   DEBUG CONSOLE   PROBLEMS   TERMINAL   PORTS

PS C:\Users\sivas\OneDrive\Desktop\Git\case> git log
    added

commit 0bc6389afe4bf79a9b92a56e60f9e2a1a5fa365a
Author: Sivashree-117 <sivashree.pl2023@vitstudent.ac.in>
Date:   Tue Dec 23 19:40:59 2025 +0530

    Update info.py

commit 854b75f611a295d124c1b0b8668ca3802d54fa28 (HEAD -> main, origin/main, origin/HEAD)
Author: Sivashree-117 <sivashree.pl2023@vitstudent.ac.in>
Date:   Tue Dec 23 19:57:41 2025 +0530

    add

commit eef25d09a81729cee55ef6ebb82085e5a33cc05b
Author: Sivashree-117 <sivashree.pl2023@vitstudent.ac.in>
Date:   Tue Dec 23 19:47:50 2025 +0530

    added

commit 0bc6389afe4bf79a9b92a56e60f9e2a1a5fa365a
Author: Sivashree-117 <sivashree.pl2023@vitstudent.ac.in>
Date:   Tue Dec 23 19:40:59 2025 +0530

    Update info.py

commit dda3f242317950949e3932657df8f7510652bf3b
Author: Sivashree-117 <sivashree.pl2023@vitstudent.ac.in>
Date:   Tue Dec 23 18:45:08 2025 +0530
:...skipping...
commit 854b75f611a295d124c1b0b8668ca3802d54fa28 (HEAD -> main, origin/main, origin/HEAD)
Author: Sivashree-117 <sivashree.pl2023@vitstudent.ac.in>
Date:   Tue Dec 23 19:57:41 2025 +0530

    add

commit eef25d09a81729cee55ef6ebb82085e5a33cc05b
Author: Sivashree-117 <sivashree.pl2023@vitstudent.ac.in>
Date:   Tue Dec 23 19:47:50 2025 +0530

    added

commit 0bc6389afe4bf79a9b92a56e60f9e2a1a5fa365a
```

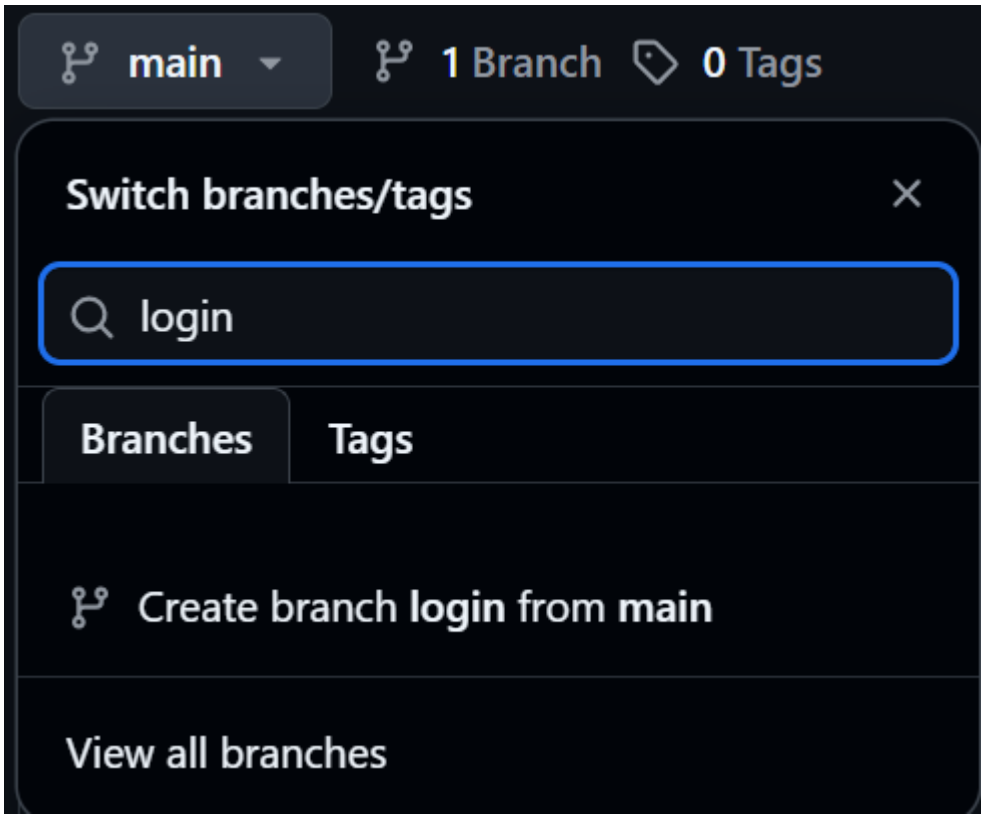
Case Study 3: Feature Development Using Branches

Scenario: A new Login feature must be developed without disturbing main code.

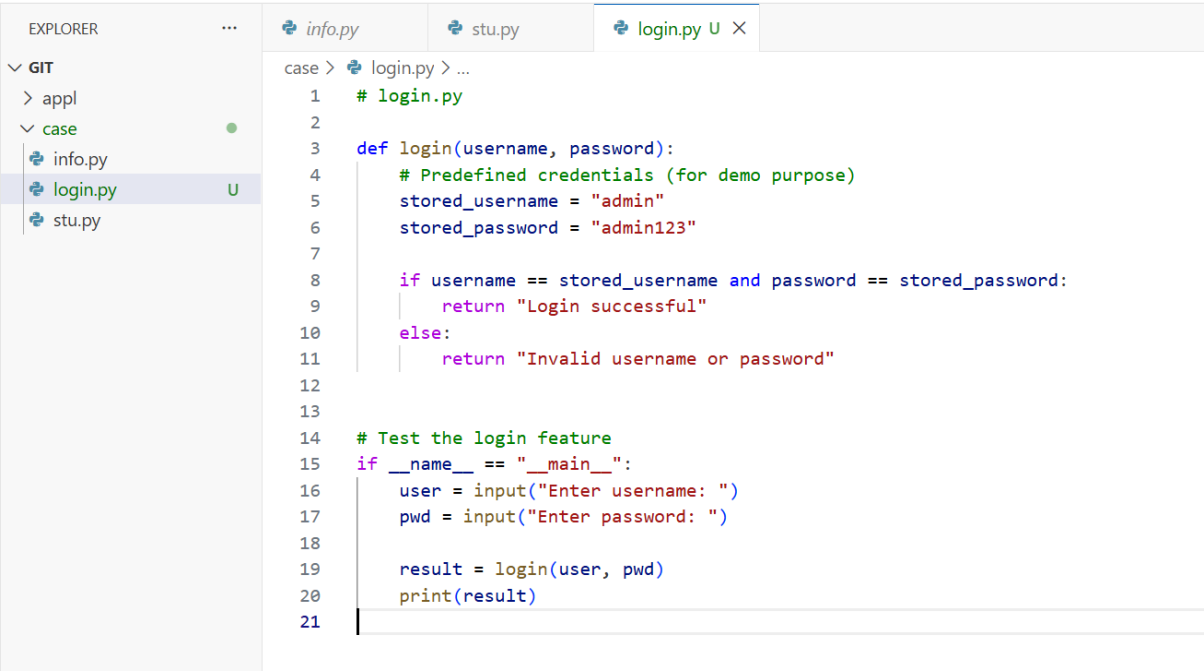
Tasks:

- Create a new branch

(Creatin new branch in git repo)



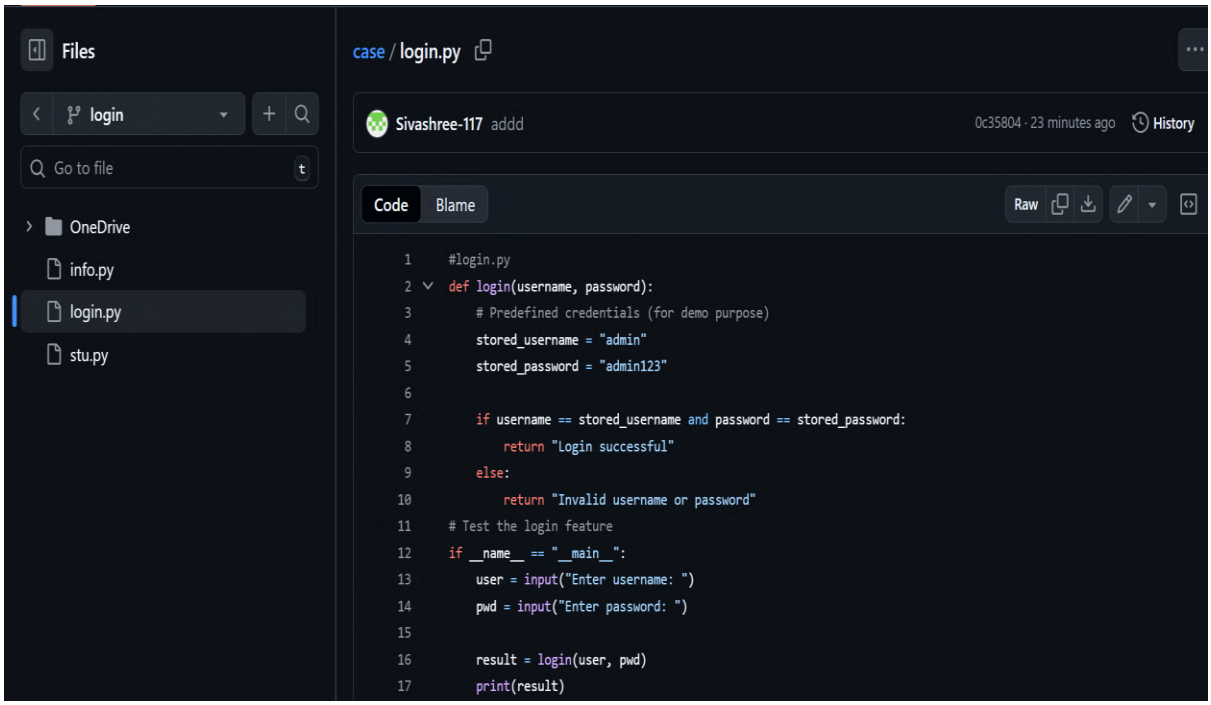
- Add login-related code



- Commit changes

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git add login.py
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git commit -m "addd"
[login 0c35804] addd
 1 file changed, 17 insertions(+)
 create mode 100644 login.py
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git push origin login
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 584 bytes | 584.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/case.git
   ff0c788..0c35804  login -> login
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
```

(Changes added to git)



- Switch between branches

→ **git branch -a**

→ **git checkout branchname**

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git branch -a
* main
  remotes/origin/HEAD -> origin/main
  remotes/origin/login
  remotes/origin/main
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git checkout login
>>
branch 'login' set up to track 'origin/login'.
Switched to a new branch 'login'
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git branch -a
* login
  main
  remotes/origin/HEAD -> origin/main
  remotes/origin/login
  remotes/origin/main
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
```

Case Study 4: Merging Feature into Main Code

Scenario: The login feature is tested and approved.

Tasks:

- Switch to main branch

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git branch -a
* login
  main
  remotes/origin/HEAD -> origin/main
  remotes/origin/login
  remotes/origin/main
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git checkout main
```

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git branch -a
  login
* main
  remotes/origin/HEAD -> origin/main
  remotes/origin/login
  remotes/origin/main
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
```

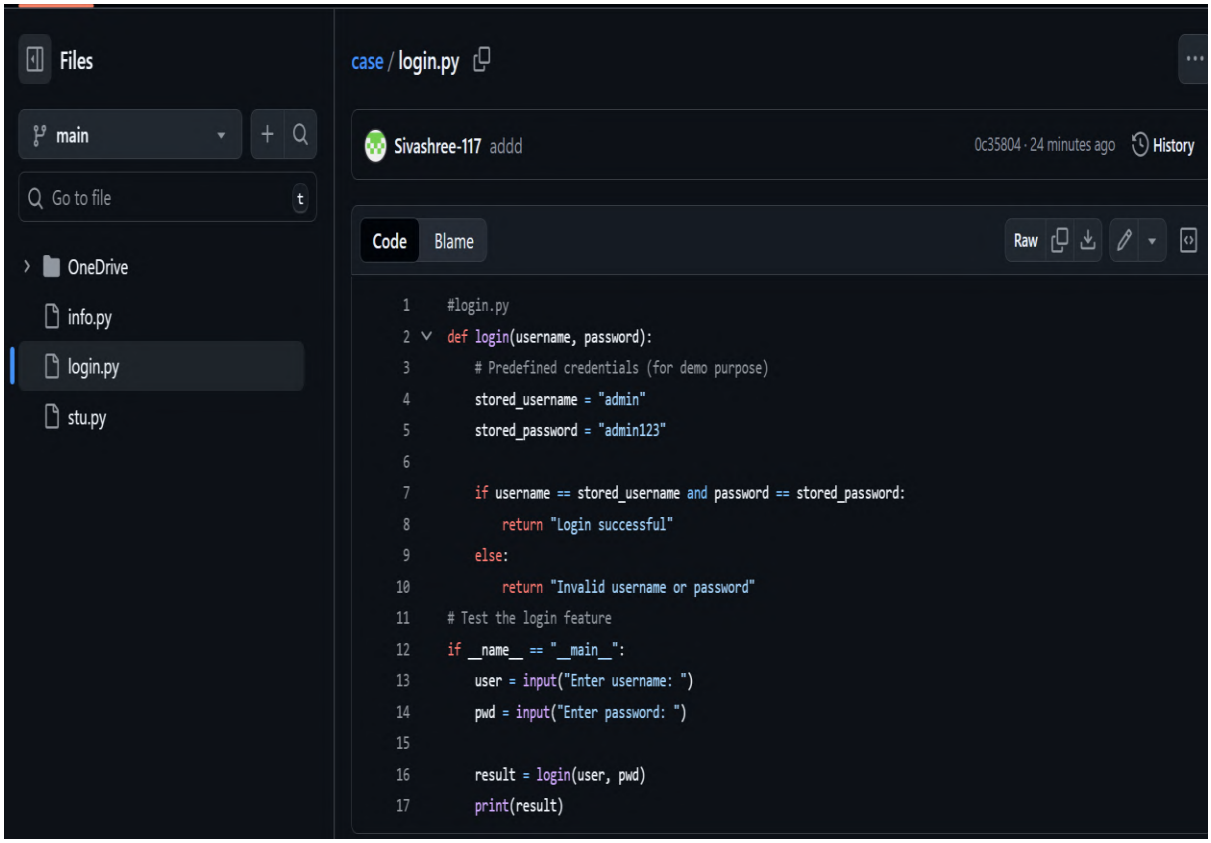
Ln 1, Col 10

- Merge feature branch

→ **git merge branchname**

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git merge login
Updating 854b75f..0c35804
Fast-forward
 OneDrive/Desktop/Git/case |  1 +
 login.py                  | 17 +++++++++++++++++
 2 files changed, 18 insertions(+)
 create mode 160000 OneDrive/Desktop/Git/case
 create mode 100644 login.py
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
```

- Verify merged content

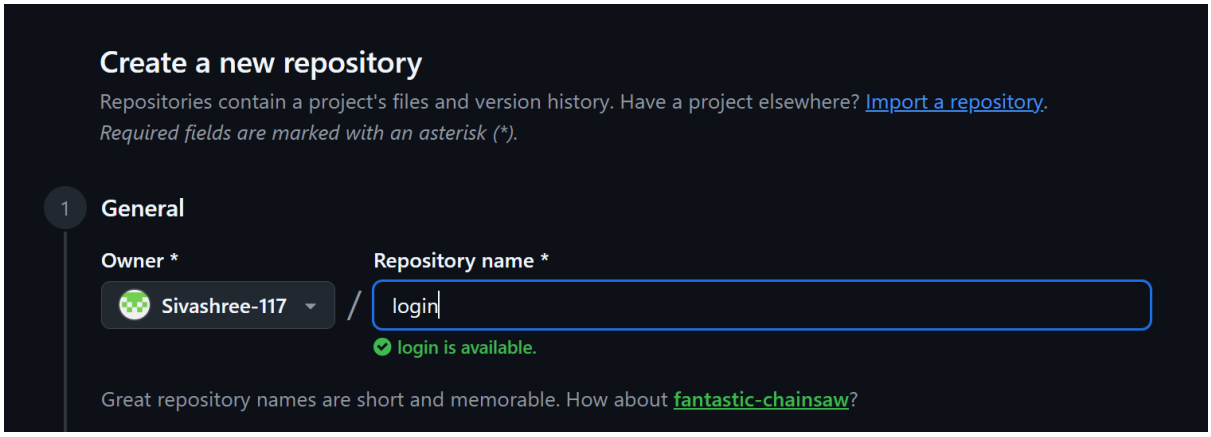


Case Study 5: Working with Remote Repository

Scenario: The project must be shared using GitHub.

Tasks:

- Create remote repository



- Add remote origin

```
PS C:\Users\sivas\OneDrive\Desktop\Git> git remote set-url origin https://gi
thub.com/Sivashree-117/login.git
>>
PS C:\Users\sivas\OneDrive\Desktop\Git>
```

- Push code to GitHub

```
PS C:\Users\sivas\OneDrive\Desktop\Git> git branch -M main
PS C:\Users\sivas\OneDrive\Desktop\Git> git push -u origin main
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 16 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (17/17), 2.73 KiB | 349.00 KiB/s, done.
Total 17 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/login.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\sivas\OneDrive\Desktop\Git>
```

Case Study 6: Undoing Mistakes

Scenario: A file is modified incorrectly.

Tasks:

- View file changes

(Old file)

case > login.py > login

```python
#login.py
def login(username, password):
    # Predefined credentials (for demo purpose)
    stored_username = "admin"
    stored_password = "admin123"

```

(Modified file – admin is changed to siva)

info.py | stu.py | login.py

case > login.py > login

```python
#login.py
def login(username, password):
    # Predefined credentials (for demo purpose)
    stored_username = "siva"
    stored_password = "siva117"
```

(Show difference between modified file and original file)

→ **git diff**

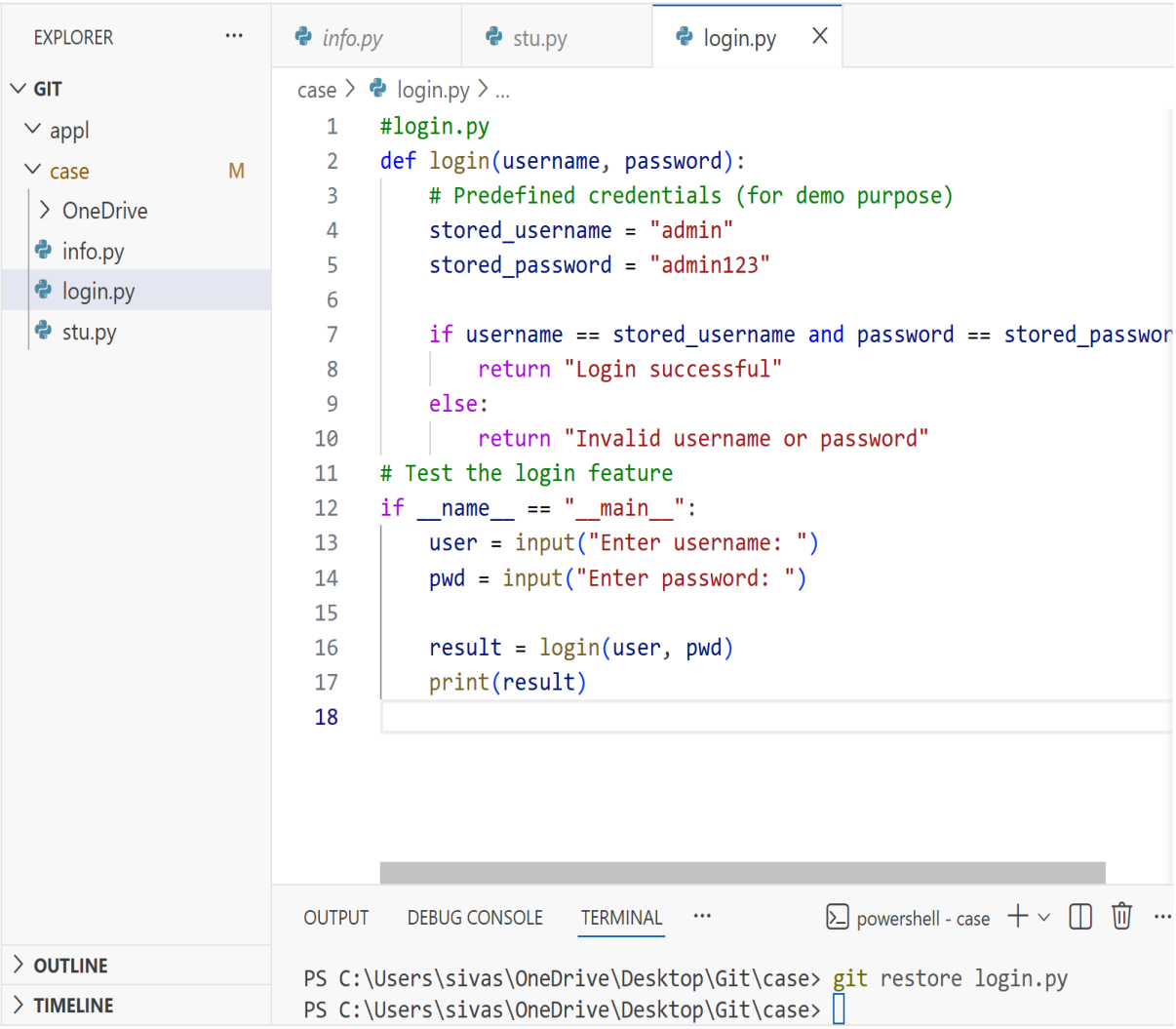
(Dirty indicates that there is a change in that file)

```
PS C:\Users\sivas\OneDrive\Desktop\Git> git diff
diff --git a/OneDrive/Desktop/Git/case b/OneDrive/Desktop/Git/case
--- a/OneDrive/Desktop/Git/case
+++ b/OneDrive/Desktop/Git/case
@@ -1 +1 @@
-Subproject commit 0c358041d946689bbbb4fa6caf3145c1254a7e3e
+Subproject commit 0c358041d946689bbbb4fa6caf3145c1254a7e3e-dirty
PS C:\Users\sivas\OneDrive\Desktop\Git>
```

- Discard uncommitted changes

→ **git restore filename**

(Discard uncommitted changes and bring back the original file)



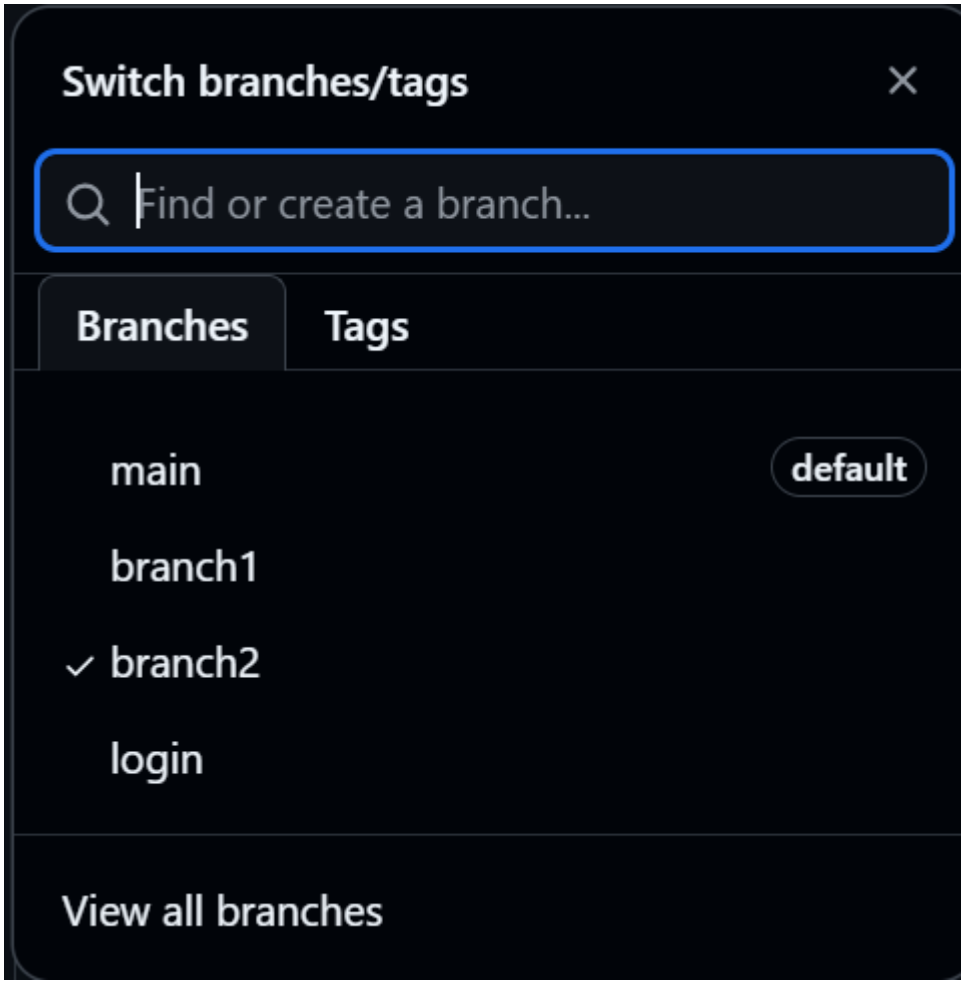
Case Study 7: Handling Merge Conflicts

Scenario: Two developers edited the same file.

Tasks:

- Merge branches

(Two new branch is created – branch1 , branch2)



(git branch -a shows available branches – main , login , branch1 , branch2)

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git branch -a
  login
* main
  remotes/origin/HEAD -> origin/main
  remotes/origin/branch1
  remotes/origin/branch2
  remotes/origin/login
  remotes/origin/main
```

(First person use branch1 and edit username as branch1 and merged to main branch similarly second person use branch2 and edit username as branch2 and when the person tried to merge a conflict is happen)

case > login.py > ...

```python
#login.py
def login(username, password):
    # Predefined credentials (for demo purpose)
    stored_username = "branch1"
    stored_password = "branch123"

    if username == stored_username and password == stored_password:
        return "Login successful"
    else:
        return "Invalid username or password"
# Test the login feature
if __name__ == "__main__":
    user = input("Enter username: ")
    pwd = input("Enter password: ")

    result = login(user, pwd)
    print(result)
```

(Conflict shown in VS code)

info.py | stu.py | login.py 9+, !

case > login.py > ...

```python
#login.py
def login(username, password):
    # Predefined credentials (for demo purpose)
Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
<<<<<<< HEAD (Current Change)
    stored_username = "branch1"
=======
    stored_username = "branch2"
>>>>>>> branch2 (Incoming Change)
    stored_password = "branch123"

    if username == stored_username and password == stored_password:
        return "Login successful"
    else:
        return "Invalid username or password"
# Test the login feature
if __name__ == "__main__":
    user = input("Enter username: ")
    pwd = input("Enter password: ")
```

Resolve in Merge Editor

OUTPUT | DEBUG CONSOLE | PROBLEMS 12 | TERMINAL | PORTS | powershell - case

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git push origin branch2
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/case.git
   0c35804..4b76ecd  branch2 -> branch2
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git merge branch2
Auto-merging login.py
CONFLICT (content): Merge conflict in login.py
Automatic merge failed; fix conflicts and then commit the result.
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
```

4 login.py

```diff
@@ -1,8 +1,8 @@
 #login.py
 def login(username, password):
     # Predefined credentials (for demo purpose)
-    stored_username = "admin"
-    stored_password = "admin123"
+    stored_username = "branch2"
+    stored_password = "branch123"

     if username == stored_username and password == stored_password:
         return "Login successful"
```

- Resolve conflicts

case > login.py > login

Incoming 4b76ecd • refs/remotes/origin/branch2, refs/head...

```
#login.py
def login(username, password):
    # Predefined credentials (for demo purpose)
    stored_username = "branch2"
    stored_password = "branch123"
```

Current f7a224f • refs/heads/main, refs/remotes/origin/mai...

```
#login.py
def login(username, password):
    # Predefined credentials (for demo purpose)
    stored_username = "branch1"
    stored_password = "branch123"
```

Result case\login.py — 0 Conflicts Remaining

```
#login.py
def login(username, password):
    # Predefined credentials (for demo purpose)
```

Manual Resolution | Reset to base

```
    stored_username = "branch2"
    stored_password = "admin123"

    if username == stored_username and password == stored_password:
        return "Login successful"
    else:
        return "Invalid username or password"
```

Complete Merge

(Changes added to vs code)

stu.py | login.py M

case > login.py > ...

```
#login.py
def login(username, password):
    # Predefined credentials (for demo purpose)
    stored_username = "branch2"
    stored_password = "admin123"

    if username == stored_username and password == stored_password:
        return "Login successful"
    else:
        return "Invalid username or password"
# Test the login feature
if __name__ == "__main__":
    user = input("Enter username: ")
    pwd = input("Enter password: ")

    result = login(user, pwd)
    print(result)
```

- Commit resolved changes

(Conflict problem is solved by entering one proper code)

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git add .
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git commit -m "addd"
[main 0ba712a] addd
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 327 bytes | 163.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Sivashree-117/case.git
   f7a224f..0ba712a  main -> main
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
```

(Now conflict is solved and only one username is shown in git repo)

case / login.py

Sivashree-117 addd

Code Blame 19 lines (15 loc) · 489 Bytes

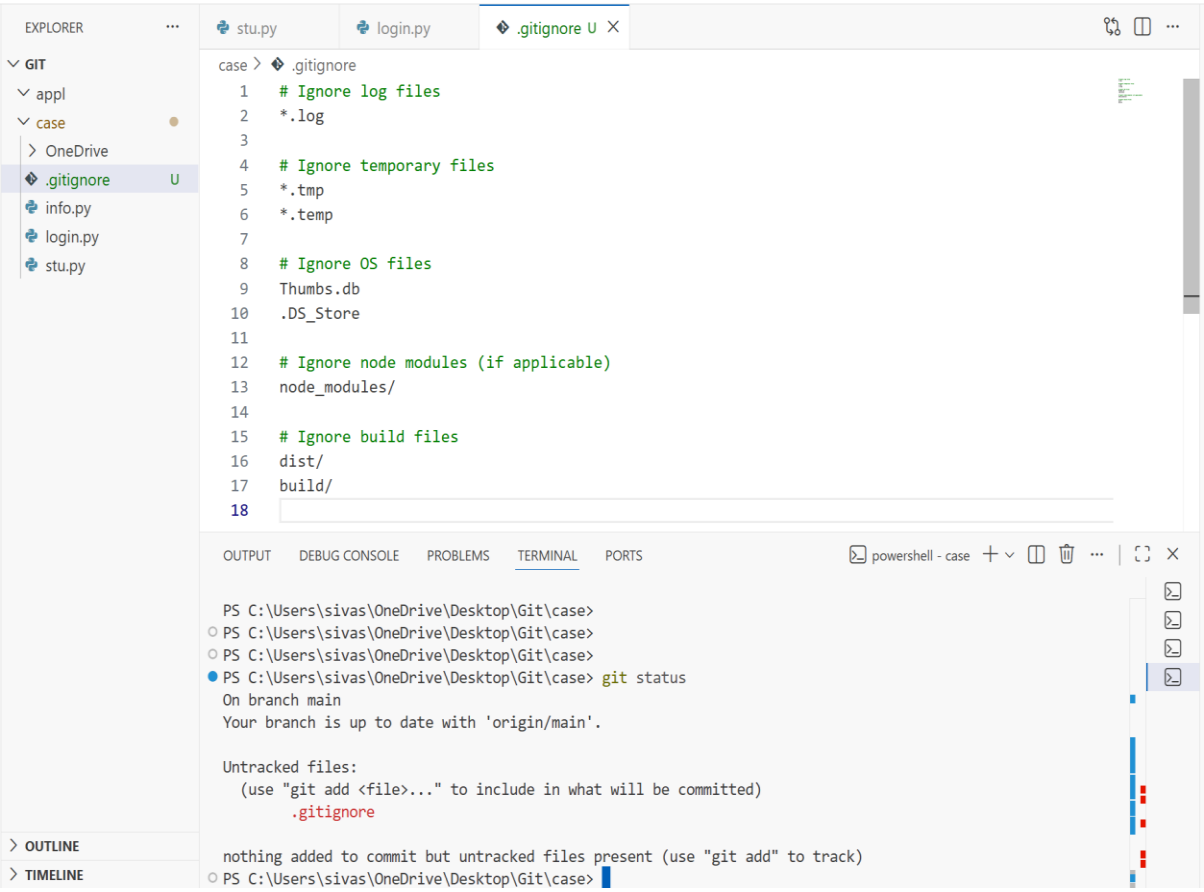
```python
#login.py
def login(username, password):
    # Predefined credentials (for demo purpose)
    stored_username = "branch2"
    stored_password = "admin123"

    if username == stored_username and password == stored_password:
        return "Login successful"
    else:
        return "Invalid username or password"
# Test the login feature
if __name__ == "__main__":
    user = input("Enter username: ")
    pwd = input("Enter password: ")

    result = login(user, pwd)
    print(result)
```

Case Study 8: Ignoring Unwanted Files

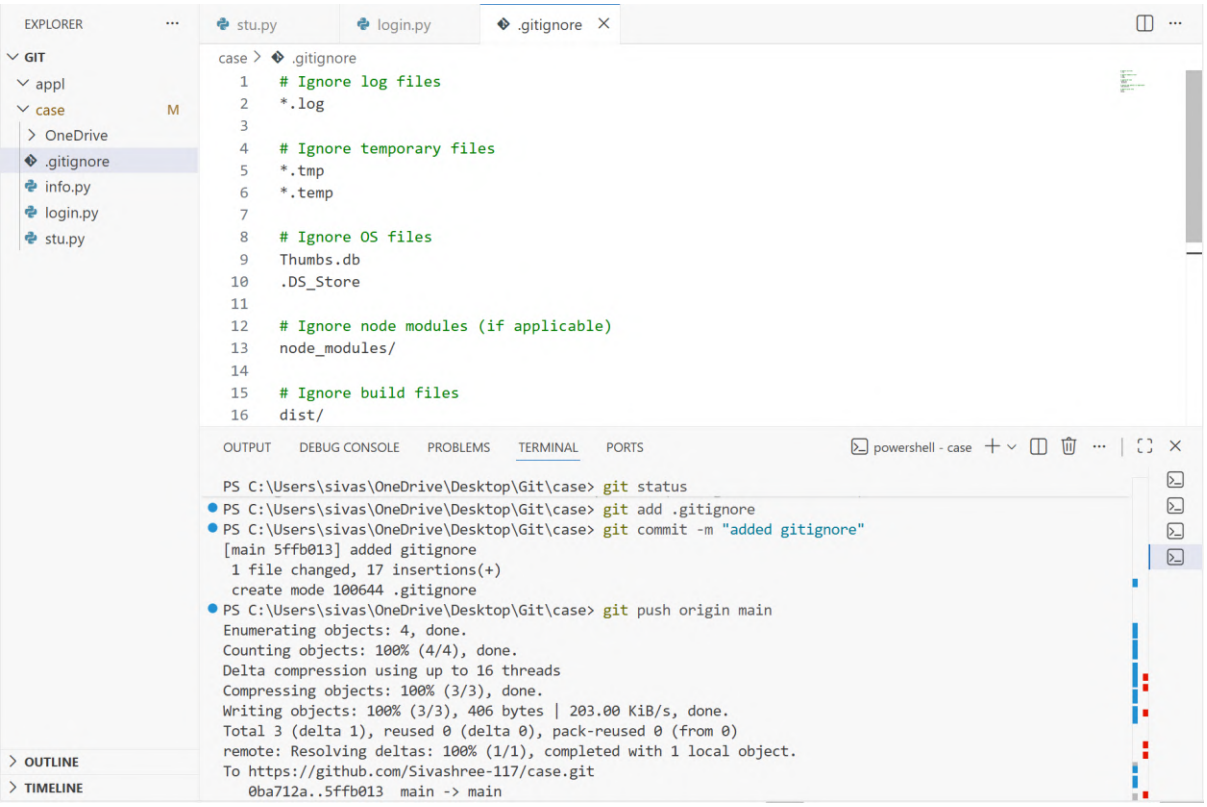
Scenario: Log and temporary files should not be tracked.

Tasks:

- Create .gitignore file



- Add ignore patterns



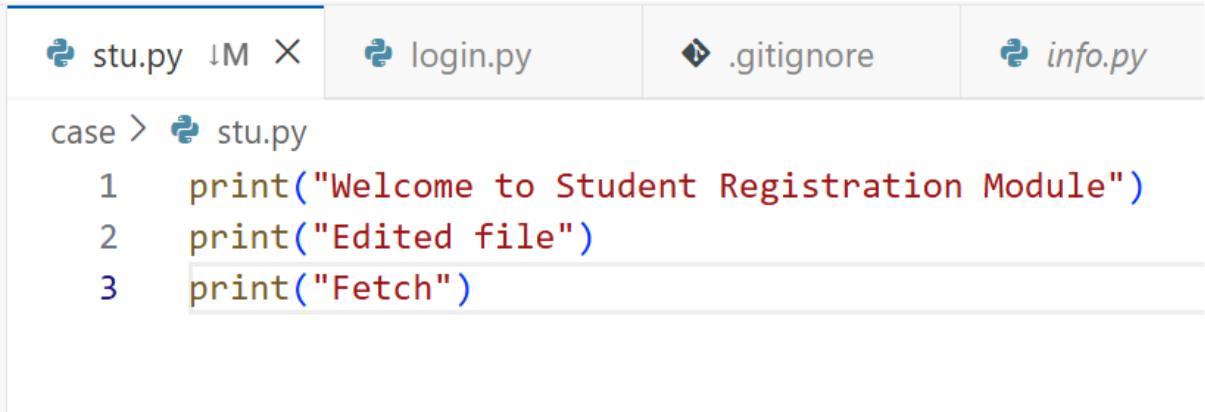
Case Study 9: Collaborating with Team

Scenario: A teammate pushed new changes.

Tasks:

- Fetch latest changes

(Actual file)



(File is edited in Git)

case / stu.py

Sivashree-117 Update stu.py

Code Blame 4 lines (4 loc) · 144 Bytes

```python
print("Welcome to Student Registration Module")
print("Edited file")
print("Fetch")
print("This file is see difference between fetch and pull")
```

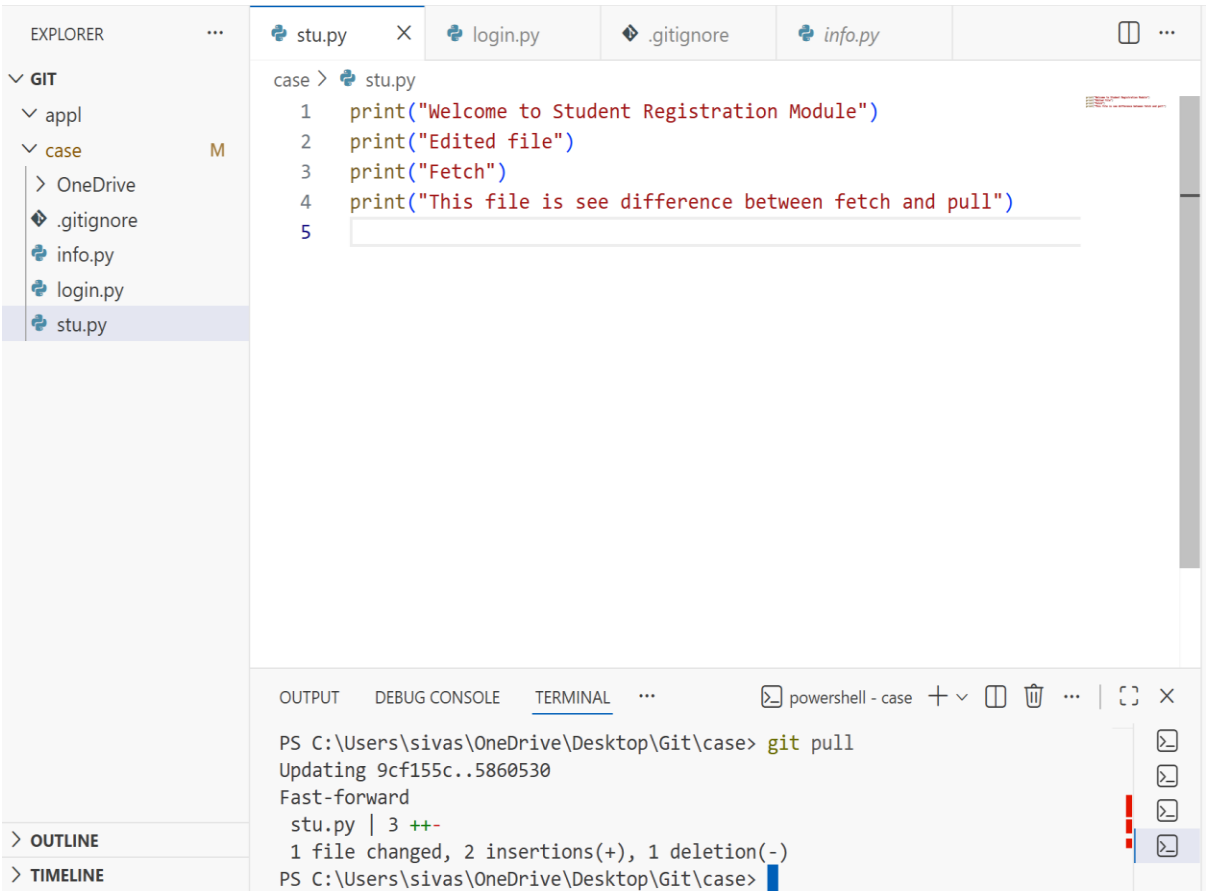
(git diff will fetch the latest change but it wont modify the code)

OUTPUT DEBUG CONSOLE TERMINAL ··· powershell - case

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git status
On branch main
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)

nothing to commit, working tree clean
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git fetch
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git fetch origin
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git diff origin/main
>>
diff --git a/stu.py b/stu.py
index aeb3e11..08d4680 100644
--- a/stu.py
+++ b/stu.py
@@ -1,4 +1,3 @@
 print("Welcome to Student Registration Module")
 print("Edited file")
-print("Fetch")
-print("This file is see difference between fetch and pull")
+print("Fetch")
```

- Pull updates

(Make changes to existing code)



- Resolve conflicts if any

No conflicts

Case Study 10: Final Project Release

Scenario:

Project is ready for final submission.

Tasks:

- Review commit history

→ **git log –oneline**

```
OUTPUT    DEBUG CONSOLE    TERMINAL    ···          powershell - case
\ No newline at end of file
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
 …
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git log --oneline
>>
5860530 (HEAD -> main, origin/main, origin/HEAD) Update stu.py
9cf155c ad
5ffb013 added gitignore
0ba712a addd
4b76ecd (origin/branch2, branch2) ad
f7a224f (origin/branch1, branch1) add
0c35804 (origin/login, login) addd
ff0c788 add1
854b75f add
eef25d0 added
0bc6389 Update info.py
dda3f24 Create info.py
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
```

- Push final code

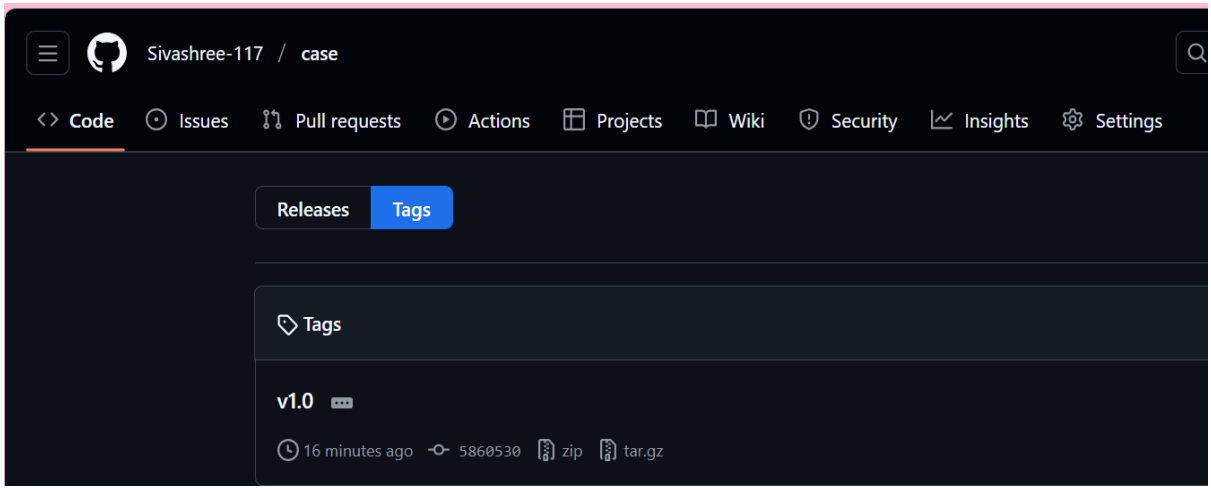
```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git branch
  branch1
  branch2
  login
* main
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git push origin main
Everything up-to-date
PS C:\Users\sivas\OneDrive\Desktop\Git\case>
```

- Prepare release version

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git tag v1.0
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git push origin v1.0
>>
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/case.git
 * [new tag]         v1.0 -> v1.0
```

```
PS C:\Users\sivas\OneDrive\Desktop\Git\case> git tag
v1.0
```

(git version is shown in git repository)



In lab Assignment :

(Question 3 and 4)

(Team members : Likitha , Kavi , Vedha , Sumathi , Siva)

Question 3:

An electricity board calculates charges based on usage.

Write a program that:

• Takes number of units consumed

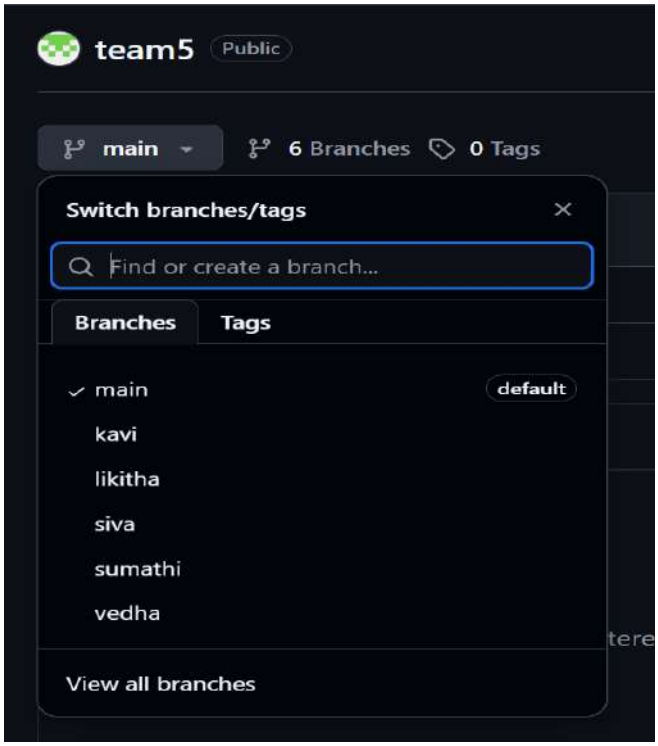
• Calculates bill:

o First 100 units → ₹2/unit

o Next 100 units → ₹3/unit

o Remaining → ₹5/unit

• Displays total bill

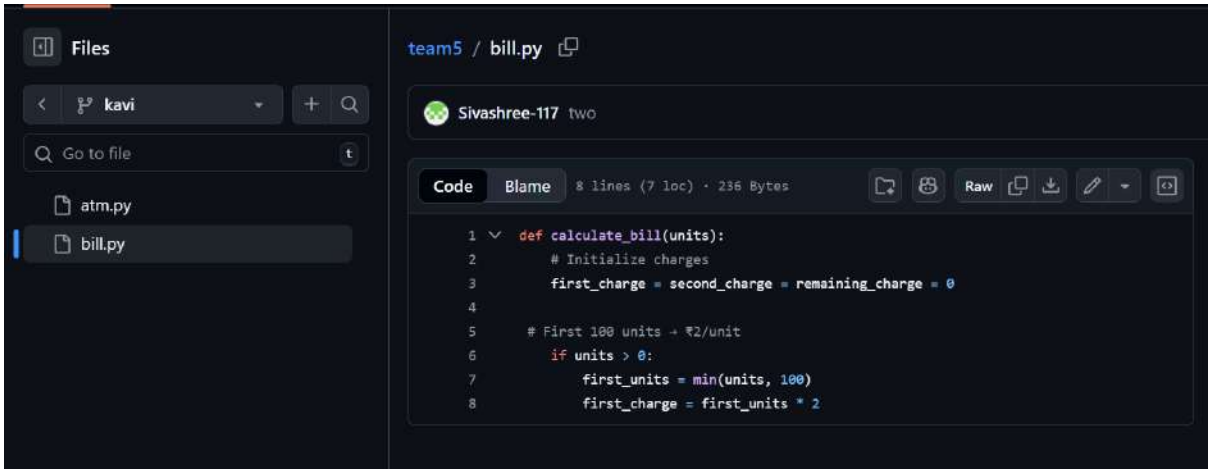



```
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git branch -a
* main
  remotes/origin/HEAD -> origin/main
  remotes/origin/kavi
  remotes/origin/likitha
  remotes/origin/main
  remotes/origin/siva
  remotes/origin/sumathi
  remotes/origin/vedha
PS C:\Users\sivas\OneDrive\Desktop\agile\team5>
```

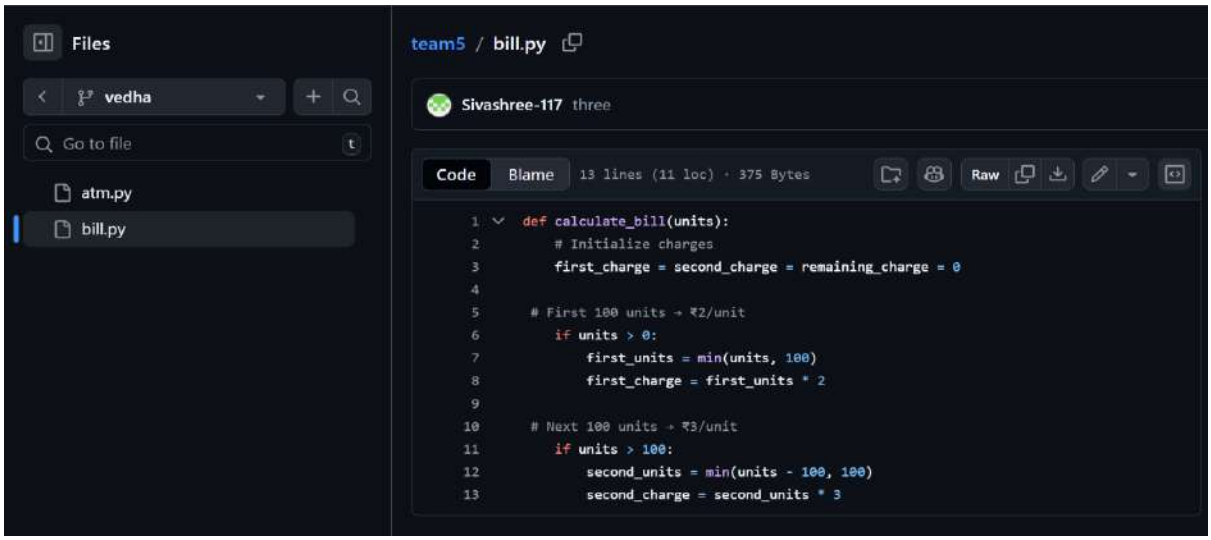
OUTPUT DEBUG CONSOLE PROBLEMS TERMINAL PORTS

```
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git branch -a
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout likitha
branch 'likitha' set up to track 'origin/likitha'.
Switched to a new branch 'likitha'
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git add .
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git commit -m "one"
[likitha d7110c1] one
 1 file changed, 3 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin likitha
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 356 bytes | 178.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/team5.git
   73bf551..d7110c1  likitha -> likitha
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git merge likitha
Updating 73bf551..d7110c1
Fast-forward
 bill.py | 4 +++-
 1 file changed, 3 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin main
fatal: unable to access 'https://github.com/Sivashree-117/team5.git/':
nect to server
PS C:\Users\sivas\OneDrive\Desktop\agile\team5>
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/team5.git
   73bf551..d7110c1  main -> main
```

```
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout kavi
branch 'kavi' set up to track 'origin/kavi'.
Switched to a new branch 'kavi'
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git pull origin main
From https://github.com/Sivashree-117/team5
 * branch            main       -> FETCH_HEAD
Updating 73bf551..d7110c1
Fast-forward
 bill.py | 4 +++-
 1 file changed, 3 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git add .
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git commit -m "two"
[kavi 31eea94] two
 1 file changed, 5 insertions(+)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git pull origin kavi
From https://github.com/Sivashree-117/team5
 * branch            kavi       -> FETCH_HEAD
Already up to date.
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin kavi
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 420 bytes | 140.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/team5.git
   73bf551..31eea94  kavi -> kavi
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git merge kavi
Updating d7110c1..31eea94
Fast-forward
 bill.py | 5 +++++
 1 file changed, 5 insertions(+)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/team5.git
   d7110c1..31eea94  main -> main
```



```
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout vedha
branch 'vedha' set up to track 'origin/vedha'.
Switched to a new branch 'vedha'
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git pull origin main
From https://github.com/Sivashree-117/team5
 * branch            main       -> FETCH_HEAD
Updating 73bf551..31eea94
Fast-forward
 bill.py | 7 +++++++
 1 file changed, 7 insertions(+)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git add .
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git commit -m "three"
[vedha eda1ab3] three
 1 file changed, 6 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin vedha
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 388 bytes | 388.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Sivashree-117/team5.git
   73bf551..eda1ab3  vedha -> vedha
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin main
Everything up-to-date
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git merge vedha
Updating 31eea94..eda1ab3
Fast-forward
 bill.py | 7 ++++++-
 1 file changed, 6 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/team5.git
   31eea94..eda1ab3  main -> main
```



```
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout sumathi
branch 'sumathi' set up to track 'origin/sumathi'.
Switched to a new branch 'sumathi'
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git pull origin main
From https://github.com/Sivashree-117/team5
 * branch            main       -> FETCH_HEAD
Updating 73bf551..eda1ab3
Fast-forward
 bill.py | 12 ++++++++++++
 1 file changed, 12 insertions(+)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git add .
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git commit -m "four"
[sumathi 957169a] four
 1 file changed, 11 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin sumathi
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 435 bytes | 145.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Sivashree-117/team5.git
   73bf551..957169a  sumathi -> sumathi
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Fast-forward
 bill.py | 12 +++++++++++-
 1 file changed, 11 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/team5.git
   eda1ab3..957169a  main -> main
```

```
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout siva
branch 'siva' set up to track 'origin/siva'.
Switched to a new branch 'siva'
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git pull origin main
fatal: unable to access 'https://github.com/Sivashree-117/team5.git/':
nect to server
PS C:\Users\sivas\OneDrive\Desktop\agile\team5>
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git pull origin main
From https://github.com/Sivashree-117/team5
 * branch            main       -> FETCH_HEAD
Updating 73bf551..957169a
Fast-forward
 bill.py | 22 ++++++++++++++++++++++
 1 file changed, 22 insertions(+)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git add .
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git commit -m "five"
[siva 05eb5c1] five
 1 file changed, 13 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin siva
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 522 bytes | 174.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Sivashree-117/team5.git
   73bf551..05eb5c1  siva -> siva
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git merge siva
Updating 957169a..05eb5c1
Fast-forward
 bill.py | 14 +++++++++++++-
 1 file changed, 13 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/team5.git
   957169a..05eb5c1  main -> main
```

Files

sumathi

Go to file

atm.py

bill.py

team5 / bill.py

Sivashree-117 four

Code Blame 23 lines (19 loc) · 630 Bytes Raw

```python
def calculate_bill(units):
    # Initialize charges
    first_charge = second_charge = remaining_charge = 0

  # First 100 units → ₹2/unit
    if units > 0:
        first_units = min(units, 100)
        first_charge = first_units * 2

  # Next 100 units → ₹3/unit
    if units > 100:
        second_units = min(units - 100, 100)
        second_charge = second_units * 3

  # Remaining units → ₹5/unit
    if units > 200:
        remaining_units = units - 200
        remaining_charge = remaining_units * 5
    else:
        remaining_units = 0

    # Total bill
    total = first_charge + second_charge + remaining_charge
```

siva

Go to file

atm.py

bill.py

Code Blame 35 lines (28 loc) · 1.01 KB Raw

```python
    if units > 0:
        first_units = min(units, 100)
        first_charge = first_units * 2

  # Next 100 units → ₹3/unit
    if units > 100:
        second_units = min(units - 100, 100)
        second_charge = second_units * 3

  # Remaining units → ₹5/unit
    if units > 200:
        remaining_units = units - 200
        remaining_charge = remaining_units * 5
    else:
        remaining_units = 0

    # Total bill
    total = first_charge + second_charge + remaining_charge

    # Display breakdown
    print(f"Units = {units}")
    print(f"First 100 units = 100 * 2 = {first_charge}")
    print(f"Second 100 units = 100 * 3 = {second_charge}")
    print(f"Remaining {remaining_units} units = {remaining_units} * 5 = {remaining_ch
    print(f"Total Bill = {total}")


# Take input manually
units = int(input("Enter number of units consumed: "))
calculate_bill(units)
```

// Main branch

Files

main

Go to file t

atm.py

bill.py

team5 / bill.py

Sivashree-117 five

Code Blame 35 lines (28 loc) · 1.01 KB Raw

```python
def calculate_bill(units):
    # Initialize charges
    first_charge = second_charge = remaining_charge = 0

    # First 100 units → ₹2/unit
    if units > 0:
        first_units = min(units, 100)
        first_charge = first_units * 2

    # Next 100 units → ₹3/unit
    if units > 100:
        second_units = min(units - 100, 100)
        second_charge = second_units * 3

    # Remaining units → ₹5/unit
    if units > 200:
        remaining_units = units - 200
        remaining_charge = remaining_units * 5
    else:
        remaining_units = 0

    # Total bill
    total = first_charge + second_charge + remaining_charge

    # Display breakdown
    print(f"Units = {units}")
    print(f"First 100 units = 100 * 2 = {first_charge}")
```

// Output

```
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> python bill.py
Enter number of units consumed: 800
Units = 800
First 100 units = 100 * 2 = 200
Second 100 units = 100 * 3 = 300
Remaining 600 units = 600 * 5 = 3000
Total Bill = 3500
PS C:\Users\sivas\OneDrive\Desktop\agile\team5>
```

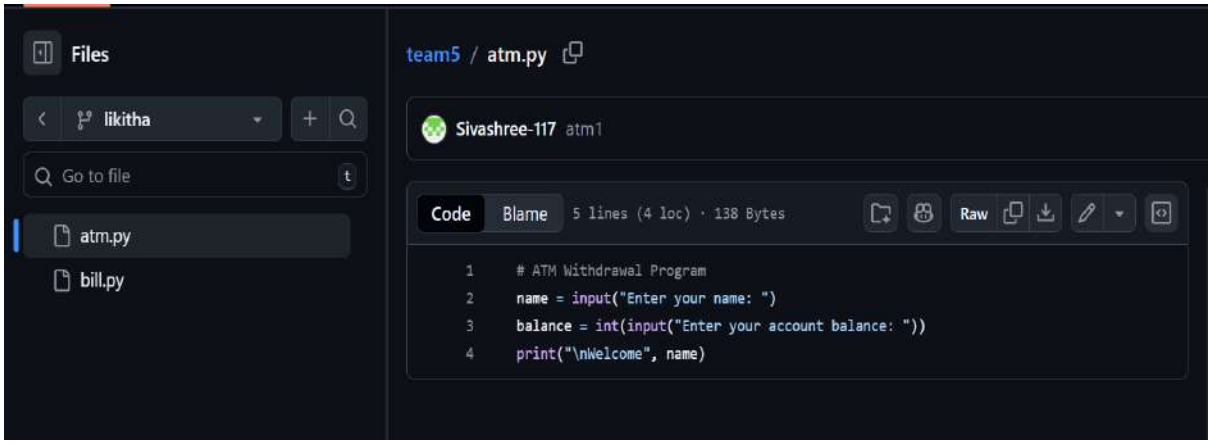
Question 4:

An ATM machine allows withdrawals only if sufficient balance is available.

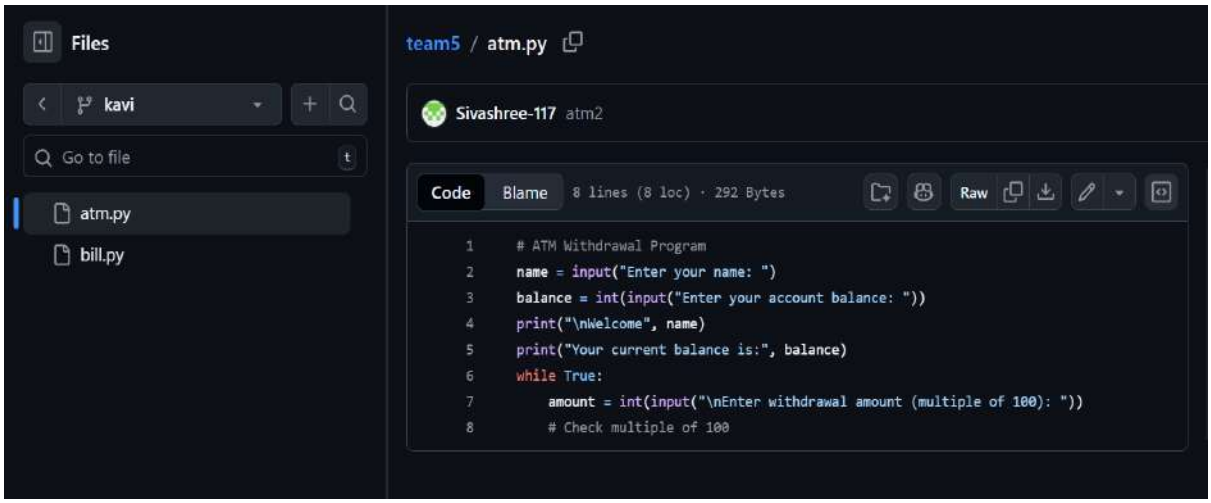
Write a program that:

- **Stores a user's balance**
- **Takes withdrawal amount as input**
- **Check**
- **. Amount must be a multiple of 100**
- **. Balance must be sufficient**
- **Deducts amount and displays updated balance**

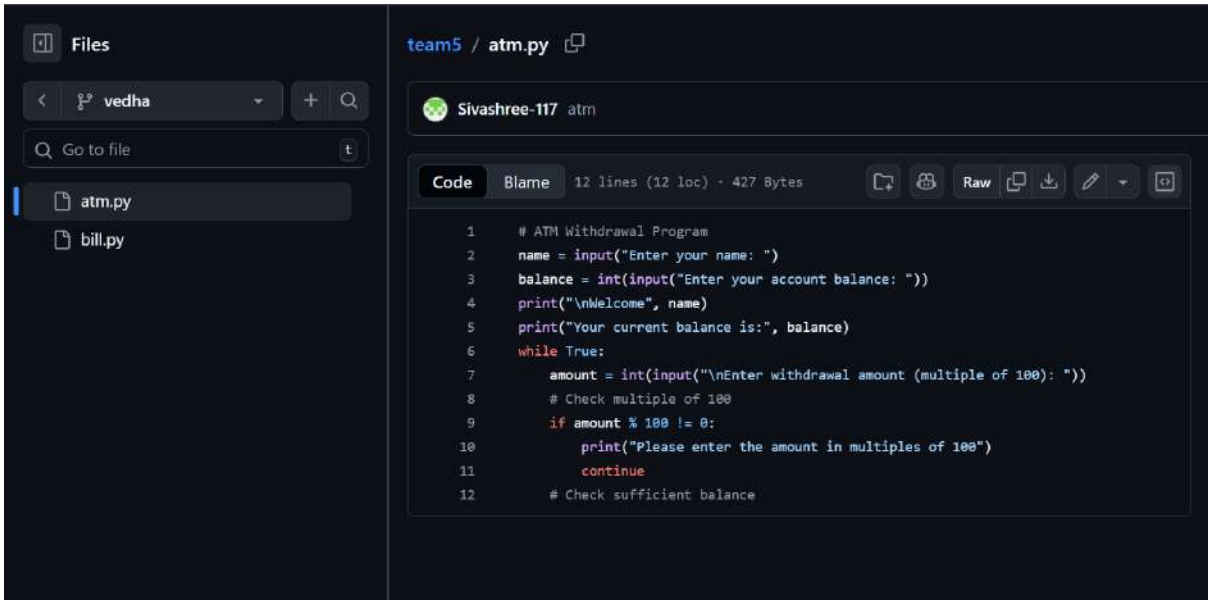
```
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout likitha
branch 'likitha' set up to track 'origin/likitha'.
Switched to a new branch 'likitha'
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git add .
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git commit -m "atm1"
[likitha ce00eb7] atm1
 1 file changed, 4 insertions(+)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin likitha
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 379 bytes | 189.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/team5.git
   fa034be..ce00eb7  likitha -> likitha
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git merge likitha
Updating fa034be..ce00eb7
Fast-forward
 atm.py | 4 ++++
 1 file changed, 4 insertions(+)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/team5.git
   fa034be..ce00eb7  main -> main
```



```
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout kavi
branch 'kavi' set up to track 'origin/kavi'.
Switched to a new branch 'kavi'
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git pull origin main
From https://github.com/Sivashree-117/team5
 * branch            main       -> FETCH_HEAD
Updating fa034be..ce00eb7
Fast-forward
 atm.py | 4 ++++
 1 file changed, 4 insertions(+)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git add .
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git commit -m "atm2"
[kavi b9ea960] atm2
 1 file changed, 4 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin kavi
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 444 bytes | 444.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/team5.git
   fa034be..b9ea960  kavi -> kavi
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git merge kavi
Updating ce00eb7..b9ea960
Fast-forward
 atm.py | 5 ++++-
 1 file changed, 4 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sivashree-117/team5.git
   ce00eb7..b9ea960  main -> main
```



```
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout vedha
branch 'vedha' set up to track 'origin/vedha'.
Switched to a new branch 'vedha'
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git pull origin main
From https://github.com/Sivashree-117/team5
 * branch            main       -> FETCH_HEAD
Updating fa034be..b9ea960
Fast-forward
 atm.py | 9 ++++++++-
 1 file changed, 8 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git add .
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git commit -m "atm3"
[vedha ae2932d] atm3
 1 file changed, 5 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git pull origin vedha
From https://github.com/Sivashree-117/team5
 * branch            vedha      -> FETCH_HEAD
Already up to date.
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git merge vedha
Updating b9ea960..ae2932d
Fast-forward
 atm.py | 6 +++++-
 1 file changed, 5 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 410 bytes | 410.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Sivashree-117/team5.git
   b9ea960..ae2932d  main -> main
```



```
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout sumathi
branch 'sumathi' set up to track 'origin/sumathi'.
Switched to a new branch 'sumathi'
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git pull origin main
From https://github.com/Sivashree-117/team5
 * branch            main       -> FETCH_HEAD
Updating fa034be..ae2932d
Fast-forward
 atm.py | 13 ++++++++++++-
 1 file changed, 12 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git add .
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git commit -m "atm4"
[sumathi 4af835a] atm4
 1 file changed, 6 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin sumathi
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 385 bytes | 385.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Sivashree-117/team5.git
   fa034be..4af835a  sumathi -> sumathi
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin main
Everything up-to-date
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git merge sumathi
Updating ae2932d..4af835a
Fast-forward
'origin/siva'.
Switched to a new branch 'siva'
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git pull origin main
From https://github.com/Sivashree-117/team5
 * branch            main       -> FETCH_HEAD
Updating fa034be..4af835a
Fast-forward
 atm.py | 18 ++++++++++++++++++-
 1 file changed, 17 insertions(+), 1 deletion(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git add .
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git commit -m "atm5"
[siva 15a47f2] atm5
 1 file changed, 9 insertions(+), 2 deletions(-)
```

Files

sumathi

Go to file

atm.py

bill.py

team5 / atm.py

Sivashree-117 atm4

Code Blame 17 lines (17 loc) · 546 Bytes Raw

```python
# ATM Withdrawal Program
name = input("Enter your name: ")
balance = int(input("Enter your account balance: "))
print("\nWelcome", name)
print("Your current balance is:", balance)
while True:
    amount = int(input("\nEnter withdrawal amount (multiple of 100): "))
    # Check multiple of 100
if amount % 100 != 0:
        print("Please enter the amount in multiples of 100")
        continue
    # Check sufficient balance
     if amount > balance:
        print("Insufficient balance")
        continue
    # Deduct amount
    balance -= amount
```

```
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout siva
M       atm.py
Switched to branch 'siva'
Your branch is ahead of 'origin/siva' by 5 commits.
  (use "git push" to publish your local commits)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git branch
  kavi
  likitha
  main
* siva
  sumathi
  vedha
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git add .
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git commit -m "atm5"
[siva 8bda325] atm5
 1 file changed, 2 insertions(+), 2 deletions(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git merge siva
Updating 15a47f2..8bda325
Fast-forward
 atm.py | 4 ++--
 1 file changed, 2 insertions(+), 2 deletions(-)
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 312 bytes | 156.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Sivashree-117/team5.git
   15a47f2..8bda325  main -> main
PS C:\Users\sivas\OneDrive\Desktop\agile\team5>
```

Files

siva

Go to file

atm.py

bill.py

team5 / atm.py

Sivashree-117 atm5

Code Blame 24 lines (24 loc) · 830 Bytes Raw

```python
# ATM Withdrawal Program
name = input("Enter your name: ")
balance = int(input("Enter your account balance: "))
print("\nWelcome", name)
print("Your current balance is:", balance)
while True:
    amount = int(input("\nEnter withdrawal amount (multiple of 100): "))
    # Check multiple of 100
    if amount % 100 != 0:
        print("Please enter the amount in multiples of 100")
        continue
    # Check sufficient balance
    if amount > balance:
        print("Insufficient balance")
        continue
    # Deduct amount
    balance -= amount
    print("Withdrawal successful!")
    print("Updated balance:", balance)
    # Ask to continue or exit
    choice = input("\nDo you want to continue or exit? (Continue/exit): ").lower()
    if choice == "exit":
        print("\nThank you for using the ATM,", name)
        break
```

//main

Files

main

Go to file

atm.py

bill.py

team5 / atm.py

Sivashree-117 atm5

Code Blame 24 lines (24 loc) · 831 Bytes Raw

```python
# ATM Withdrawal Program
name = input("Enter your name: ")
balance = int(input("Enter your account balance: "))
print("\nWelcome", name)
print("Your current balance is:", balance)
while True:
    amount = int(input("\nEnter withdrawal amount (multiple of 100): "))
    # Check multiple of 100
    if amount % 100 != 0:
        print("Please enter the amount in multiples of 100")
        continue
    # Check sufficient balance
     if amount > balance:
        print("Insufficient balance")
        continue
    # Deduct amount
    balance -= amount
    print("Withdrawal successful!")
    print("Updated balance:", balance)
    # Ask to continue or exit
    choice = input("\nDo you want to continue or exit? (continue/exit): ").lower()
    if choice == "exit":
        print("\nThank you for using the ATM,", name)
        break
```

// output

```
PS C:\Users\sivas\OneDrive\Desktop\agile\team5> python atm.py
Enter your name: Agile team
Enter your account balance: 10000

Welcome Agile team
Your current balance is: 10000

Enter withdrawal amount (multiple of 100): 117
Please enter the amount in multiples of 100

Enter withdrawal amount (multiple of 100): 3000
Withdrawal successful!
Updated balance: 7000

Do you want to continue or exit? (continue/exit): continue

Enter withdrawal amount (multiple of 100): 2000
Withdrawal successful!
Updated balance: 5000

Do you want to continue or exit? (continue/exit): exit

Thank you for using the ATM, Agile team
PS C:\Users\sivas\OneDrive\Desktop\agile\team5>
```